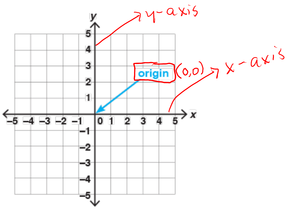
Using a Table of Points

How do you give directions? One way is to describe the path the person should take. "Drive three blocks north and six blocks west." Another way is to name major cross streets. "City Hall is near the corner of Fifth Street and Lincoln Avenue." Both methods work because cities are often organized using a grid of intersecting streets.

A **rectangular coordinate plane** works the same way. A coordinate plane, or grid, is made up of two perpendicular number lines that intersect at zero. The horizontal line is the ***x*-axis**, and the vertical line is the ***y*-axis**. The point where the lines intersect is called the **origin.** Markings at even intervals measure the distance from the origin.



You can describe a point by using the numbers along the *x*-axis and *y*-axis.

First, line the point up with a value on the *x*-axis. This is called the ***x*-coordinate**. Points to the right of the origin have a positive *x*-coordinate. Points to the left of the origin have a negative *x*-coordinate.

Then, line the point up with a value on the *y*-axis. This is called the ***y*-coordinate**. Points above the origin have a positive *y*-coordinate. Points below the origin have a negative *y*-coordinate.

Together, these two coordinates form an **ordered pair**. Ordered pairs are always written in the order (*x*, *y*). The origin is written as (0, 0).

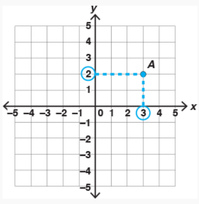
Example 1

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What are the coordinates of point *A*?

Step 1
Find the *x*-value.
Line up the point with the *x*-axis. Point *A* lines up with 3. The *x*-coordinate is 3.

Step 2
Now line up the point with the *y*-axis. Point *A* lines up with 2.
The *y*-coordinate is 2.



Step 3
Write the ordered pair. The coordinates of point *A* are (3, 2).

The *x*-axis and the *y*-axis divide the coordinate plane into four quadrants. You can determine which quadrant a point is in by looking at the signs of the coordinates.

Points with a positive *x*-coordinate and a positive *y*-coordinate are in Quadrant I.

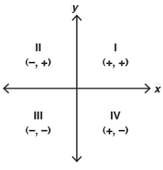
Points with a negative *x*-coordinate and a positive *y*-coordinate are in Quadrant II.

Points with a negative *x*-coordinate and a negative *y*-coordinate are in Quadrant III.

Points with a positive *x*-coordinate and a negative *y*-coordinate are in Quadrant IV.

Points that lie on an axis line are not in a quadrant.

The "grid" is also called the cartesian plane.



Equations with two variables have an infinite number of solutions. Think about the equation $y = x - 1$. For every value of *x*, there is a value of *y* that makes the equation true. For example, if $x = 5$, then $y = 4$. If $x = 3$, then $y = 2$.

$y = 5 - 1 = 4$

$y = 3 - 1 = 2$

You can use a table to organize the solutions to an equation. Then, the solutions can be used to plot points on a graph. When a line is drawn through the points, the line represents all the possible solutions to the equation.

Example 2

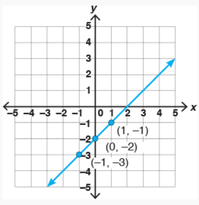
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Graph the equation $y = x - 2$.

Step 1
Make a table of values. Choose values for *x*, substitute, and solve for *y*.
If $x = -1$, then $y = -3$.
If $x = 0$, then $y = -2$.
If $x = 1$, then $y = -1$.
You can organize the work using a table. Notice how the values on the table become coordinates for points.

x	*y*	points
−1	−3	(−1, −3)
0	−2	(0, −2)
1	−1	(1, −1)

Step 2
Plot the points on the coordinate grid. Then draw a line through the points.



You could choose other *x*-values to graph more points on the line. There are an infinite number of points that will make the equation true. The line shows that the solutions go on and on. This kind of equation is called a **linear equation** because its solutions can be represented by a straight line.

Key Terms

- **linear equation**
- ***x*-axis**
- ***x*-intercept**
- ***y*-coordinate**
- **ordered pair**
- ***x*-coordinate**
- ***y*-axis**
- ***y*-intercept**

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Lesson 1: Graphing a Linear Equation

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